Full-bleed Sentinel-2 still of the solar blocks on the salt crust goes here:
`images/title_satellite.png`

The Khavda Renewable Energy Park

Engineering, industrial organisation, grid physics and political economy

$24^\circ 07'$ N, $69^\circ 21'$ E · Great Rann of Kutch, Gujarat

726 km² footprint | **30 GW** target | **Rs 1,50,000 crore** capex

{X}KDR

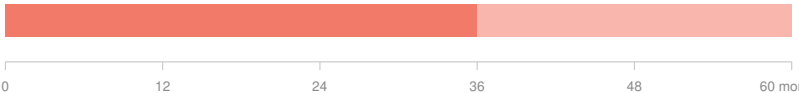
The core thesis: the two clocks

- **On-site generation clock, 12 to 18 months.** Flat state wasteland, a vertically integrated supply chain and robotic installation let developers add over 5 GW of solar and wind a year.
- **Off-site transmission clock, 36 to 60 months.** 765 kV AC and ±800 kV HVDC lines cross hundreds of kilometres of private farmland and meet right-of-way resistance.
- **The bottleneck.** By mid-2026, 14.8 GW was connected but the grid operator capped evacuation at 9,000 MW.



On-site generation clock: 12 to 18 months. Flat state wasteland, kit from Mundra, robotic install: 5 GW a year.

Off-site transmission clock: 36 to 60 months. 765 kV AC and ±800 kV HVDC across private farmland: right-of-way fights at 229 of 690 towers.



Result, mid-2026: 14.8 GW connected, 9,000 MW allowed out.

Spatial scale and how it grew

Sentinel-2 still, early 2023: salt flat, roads, first arrays

`images/sat_2023.png`

Sentinel-2 still, late 2026: dark arrays across the salt

`images/sat_2026.png`

Annotated satellite map: Pakistan border, India Bridge, KPS1, KPS2, KPS3

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72,600 ha

726 km^2 of saline mudflat, about the size of Singapore

538 km^2

held by Adani Green (53,800 ha), five times Paris

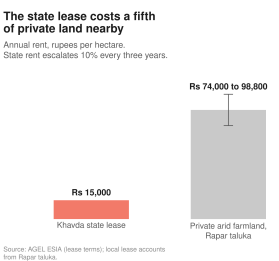
1 km

from the Pakistan border near Vighakot; Bhuj is 125 km south

Political economy of land: the 40-year wasteland lease

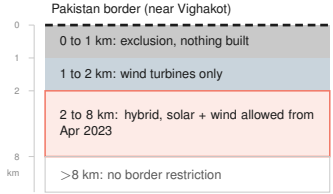
- **Around the LARR Act, 2013.** Gujarat allotted state-owned, unsurveyed revenue wasteland: no private acquisition, no litigation.
- **Tenure.** A 40-year master lease through Gujarat Power Corporation (GPCL): ten years to build, thirty to run.
- **Cost.** Rs 15,000 per hectare a year, rising 10% every three years, against Rs 74,000 to 98,800 for private arid land in Rapar taluka.
- **Anti-hoarding.** 50% of capacity within three years of the approach road, 100% within five.





Border rules and the 2023 reallocation

- **2020 allocation.** Private developers capped at 20,000 ha. SECI, a central agency, held the 23,000 ha nearest the border as a wind-only zone.
- **21 April 2023.** A South Block meeting relaxed defence norms: wind turbines 1 to 2 km from the border, hybrid solar-wind 2 to 8 km.
- **Capacity jump.** Solar beside wind lifted the parcel from 1,100 MW to 10,000 MW.
- **Pre-emptive reallocation.** Adani applied on 3 July 2023, two weeks before SECI surrendered the parcel; the state reassigned it to Adani on 24 August without a fresh tender.



Hybrid rule lifted the parcel from 1,100 MW to 10,000 MW.

Apr 2021: SECI allotted 23,000 ha next to the border, wind only

21 Apr 2023: MoD relaxes border norms (South Block meeting)

3 Jul 2023: Adani applies for the parcel

17 Jul 2023: SECI surrenders the parcel

24 Aug 2023: High-Powered Committee reallots it to Adani, no tender

A multi-developer enclave: who builds inside Khavda

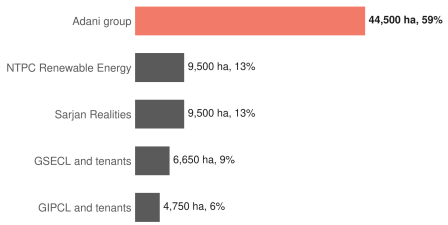
Not one firm’s park: a private anchor beside central and state public-sector utilities.

Developer	Land, ha	Target MW	Connected, late 2026
Adani Green and group	44,500–53,800	30,000	9,400+ MW; 6.6 GWh BESS
NTPC Renewable Energy	9,500–9,611	4,750	2,560–2,650 MW
Sarjan Realities (Suzlon)	9,500	4,750	1,098 MW at KPS3
GSECL and tenants	6,650	3,325	1,000–1,600 MW
GIPCL and tenants	4,750	2,375	600 MW
SECI, surrendered 2023	23,000 → Adani	3,000 wind	none

Analysis note; WRPC report (May 2026); Scroll RTI records; GIPCL.

Adani holds more land than everyone else combined

Hectares allotted, after SECI’s 23,000 ha parcel passed to Adani in 2023.



Source: analysis note; Scroll (RTI records). AGEL reports 53,800 ha including 9,500 ha leased from third parties.

Two institutional models: anchor IPP and plug-and-play landlords

- **Self-build.** AGEL and NTPC put plants directly on their parcels with in-house or contracted EPC teams.
- **Landlords.** GSECL and GIPCL act as Solar Power Park Developers: they build the 33/400 kV pooling stations, roads and water points, then sub-lease plots at 1.85 ha per MW to GUVNL auction winners. The tenants are the control group: same land, same rent, bought at auction.

Self-build: Adani, NTPC

AGEL master lease (44,500 to 53,800 ha)

↓

Step-down SPVs, one per 100 to 500 MW tranche

↓

Captive EPC; panels, turbines, cement from the group (Mundra)

Plug-and-play landlords: GSECL, GIPCL

Sub-park infrastructure: 33/400 kV pooling stations, roads, water

↓

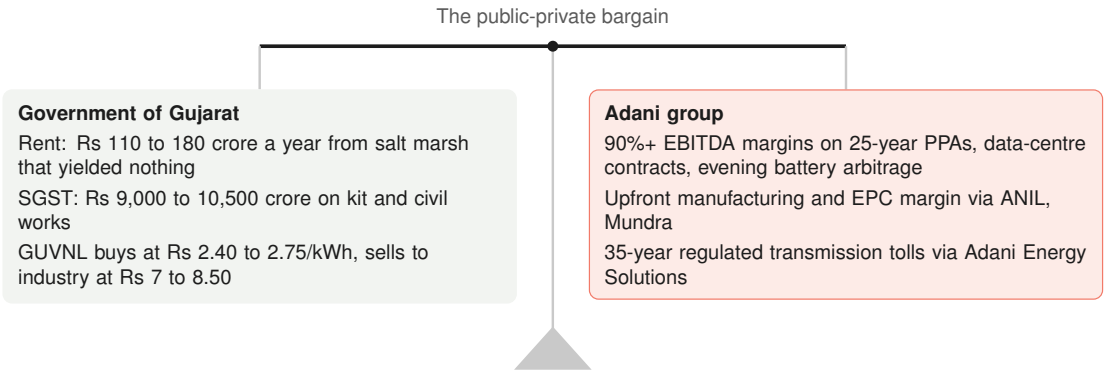
Plots sub-leased at 1.85 ha per MW to GUVNL auction winners

↓

Tenants: SAEL 1,200 MW, SJVN 1,300, NLC 600, NHPC 400, Coal India 300, JSW 300, KPI Green 200

The equilibrium: how both sides make money

The public-private bargain: barren salt marsh becomes a tax base and cheap power for the state, and a multi-stage monetisation machine for the group.

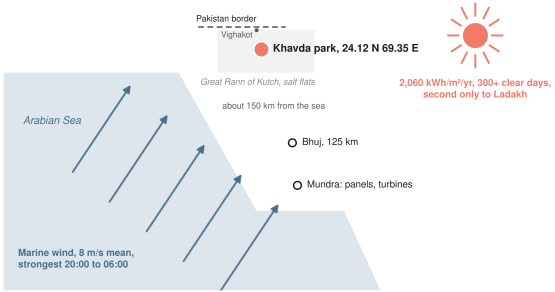


Resource physics: why Kutch is a meteorological sweet spot

- **Sun.** 2,060 kWh/m^2 a year, over 300 clear days: second in India only to Ladakh.
- **Wind.** 8 m/s on average, channelled across the flat Rann from the Arabian Sea about 150 km away.
- **Density.** 2.0 ha per MW for hybrid parks, 1.85 for pure solar sub-parks; about 2,400 modules per MW.
- **Yield.** 81 to 87 billion kWh a year across 30 GW; 1.625 GWh per hectare a year on AGEL's footprint.

Sun by day, sea wind by night, flat state land in between

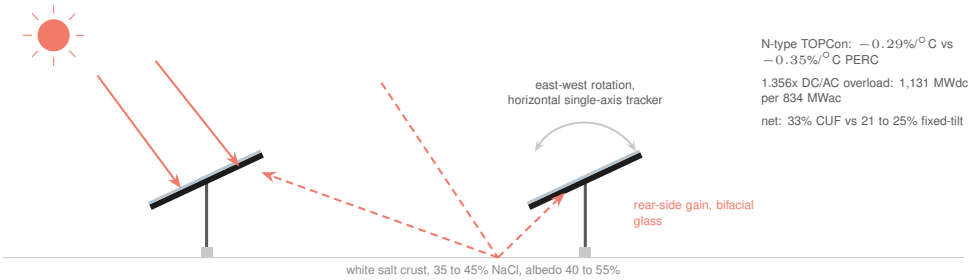
Schematic; positions approximate.



Source: AGEL project page; analysis note.

Solar hardware: N-type TOPCon, salt albedo and trackers

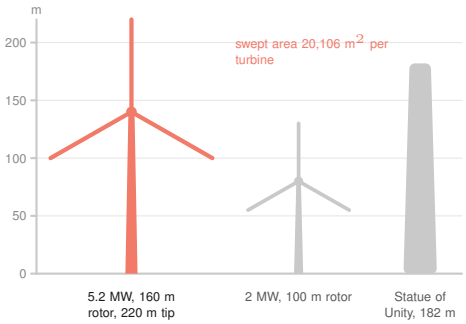
- **Cell chemistry.** Phosphorus-doped TOPCon cells avoid boron-oxygen light-induced degradation and lose less in 48 °C heat: -0.29% per °C against -0.35% for PERC.
- **Albedo.** The crust is 35 to 45% sodium chloride and reflects 40 to 55% of light onto the rear glass of bifacial modules.
- **Trackers and overload.** Horizontal single-axis trackers with a 1.356× DC-to-AC ratio (1,131 MW_{DC} per 834 MW_{AC} in KPI Green's EPC filing). Net: 33% CUF against 21 to 25% for fixed-tilt monofacial plants.



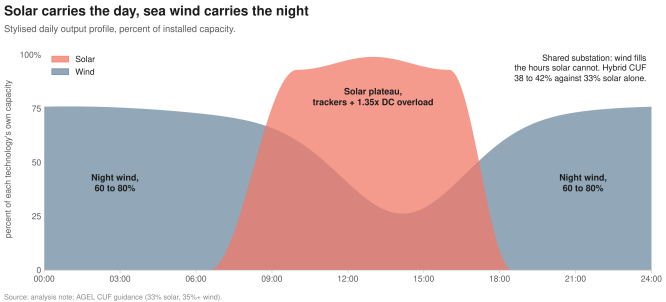
Wind aerodynamics: the 5.2 MW turbine

- **India's largest onshore turbine.** 5.2 MW, co-developed by AGEL with Wind2Energy (W2E) of Germany.
- **Dimensions.** 160 m rotor (78.6 m blades), 120 to 140 m hub, 200 to 220 m to the tip; 20,106 m^2 swept per turbine.
- **Cubic physics.** $P = \frac{1}{2}\,\rho\,A\,v^3\,C_p$. A hub at 140 m sits in faster, smoother air; in an 8 m/s regime that holds wind CUF above 35%.

A fall from 8 to 6 m/s (-25%) cuts output by 58%: siting and hub height are the whole game.



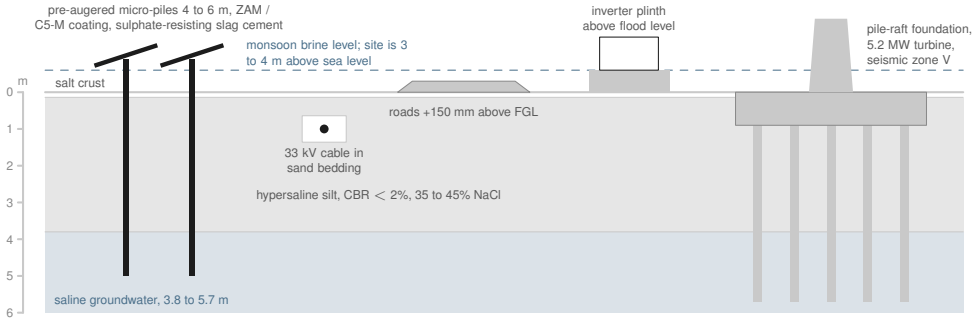
Diurnal and seasonal complementarity



- **Day, 10:00 to 16:00.** Trackers and 1.35× DC overload hold solar at a flat 90 to 100% of inverter capacity.
- **Night, 20:00 to 06:00.** The desert floor cools, marine winds accelerate: wind runs at 60 to 80% when solar is zero.
- **Monsoon.** Wind peaks June to September, when cloud cuts solar.
- **Shared substations** lift combined hybrid CUF to 38 to 42%.

Desert civil engineering: sabkha soil, brine and seismic zone V

- **Terrain.** 3 to 4 m above sea level; monsoon tides and runoff turn the top soil into hypersaline slush (CBR under 2%). Groundwater at 3.8 to 5.7 m is saline; the 2001 Bhuj earthquake (M_w 7.7) was nearby.
- **Countermeasures.** Pre-augered micro-piles 4 to 6 m deep with ZAM / C5-M coatings and sulphate-resisting slag cement; pile-raft foundations under the turbines; roads at least 150 mm above finished ground; inverters on plinths; 33 kV cables bedded in sand.

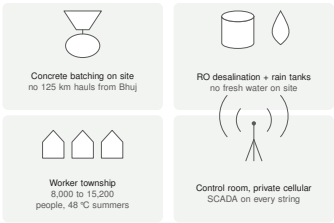


The auxiliary ecosystem and waterless operations

- **Robotic dry cleaning.** Panels are brushed three to four nights a week by IoT robots, against 50 to 70 dust storms a year, saving about 45 billion litres of water annually.
- **Payback.** Rs 12 to 15 lakh per MW of robots replaces Rs 3 to 5 lakh per MW a year of water hauling: three to four years. KPI Green alone runs 550 robots.
- **A self-contained desert city.** On-site batching plants (no 125 km hauls from Bhuj); RO desalination, rain tanks and sewage treatment; private cellular towers, fire stations and townships for 8,000 to 15,200 workers in 48 ℃ summers.

Photo: a waterless cleaning robot on a tilted tracker row

`images/robot.jpg`



The master buildout scoreboard (CMIE CapEx data)

Generation is mostly built; storage is half built; the grid spine is almost entirely still to come.

	Completed	Under implementation	Announced / pipeline
Generation, 33,425 MW across 132 projects	**12,248 MW**, 84 projects	19,057 MW, 44 projects	2,145 MW Khavda-specific
by technology	9,840 solar, 1,424 wind, 984 hybrid	14,671 solar, 436 wind, 3,950 hybrid	plus 16,000 MW multi-site (Tata, Serentica)
Battery storage, 10,296 MWh across 6 projects	**3,266 MWh**, 4 Adani SPVs	3,530 MWh (6.63 GWh energised by Oct 2026)	3,500 MWh
ISTS transmission, Rs 55,552 crore, 8 projects	**1,500 MVA**, Rs 334 crore (ICT at KPS3)	40,500 MVA, 5,616 circuit-km, Rs 55,218 crore	

37%
of tracked generation MW complete

32%
of tracked battery MWh complete

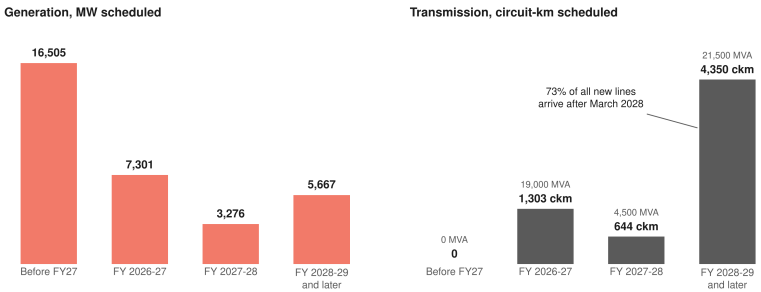
0.6%
of transmission money spent to completion

CMIE CapEx extract, November 2026. Park-level umbrella records removed; a Karnataka line in a Khavda-named SPV dropped.

Fiscal-year phasing: the two clocks in the data

The panels are scheduled years before the wires: the two clocks in CMIE's data

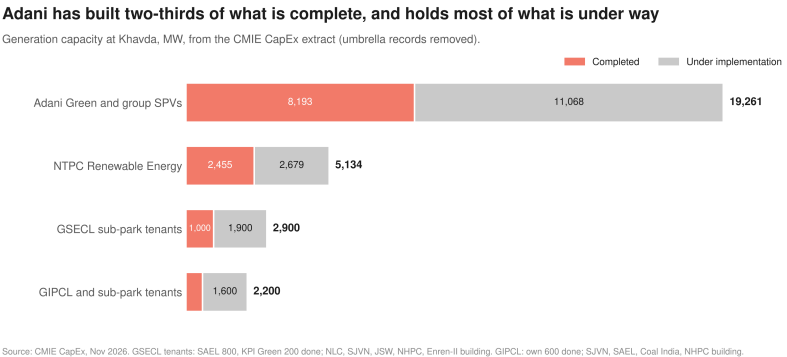
Khavda generation versus Phase IV-V transmission by scheduled completion, from the CMIE CapEx extract.



Source: CMIE CapEx, Nov 2026; 157 Khavda projects.

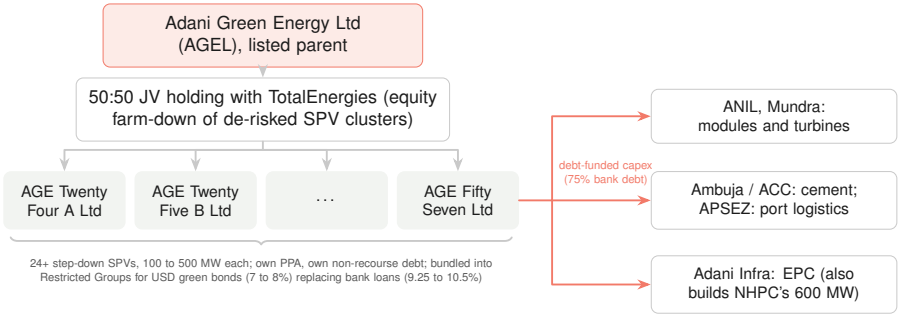
16,505 MW of generation was scheduled before FY27 against zero Phase IV–V circuit-km; 73% of the new lines (4,350 ckm, 21,500 MVA of HVDC bipoles) arrive in FY29 or later.

Developer-by-developer execution (CMIE CapEx data)



- **Adani:** 8,193 MW complete in 79 records, 11,068 MW under way, in slices of 12.5 to 362.5 MW at a filed Rs 6.5 crore/MW solar and 6.0 wind; 14 SPV projects (2,784 MW) are tied to SECI's manufacturing-linked tender.
- **NTPC:** Khavda-I 1,255 MW and Khavda-II 1,200 MW done, 2,679 MW under way. **State sub-parks:** 1,600 MW done, 3,500 MW under way across SJVN, NLC, NHPC, SAEL, Coal India, JSW and Enren-II.

Coasean industrial organisation: why Adani uses dozens of SPVs



1. Ring-fencing. Each 100 to 500 MW SPV holds its own PPA and non-recourse debt; a default or curtailment in one cannot cross-default the parent.

2. Restricted-group refinancing. Operating SPVs with 12+ months of cash flow are bundled to issue USD green bonds at 7 to 8%, retiring construction loans at 9.25 to 10.5%.

3. Equity farm-downs. 50% stakes in de-risked clusters sold to TotalEnergies recycle equity without diluting parent control.

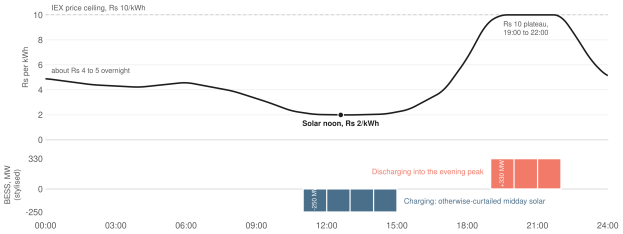
4. Vertical integration. 75% bank debt buys modules and turbines from ANIL (Mundra), cement from Ambuja/ACC, logistics from APSEZ and EPC from Adani Infra, which also builds NHPC's 600 MW block.

Revenue mechanics and the battery arbitrage pivot

- **Tier 1, 75 to 80%: 25-year PPAs.** SECI at Rs 2.92/kWh (manufacturing-linked), MSEDCL 5,000 MW at Rs 2.70, GUVNL.
- **Tier 2, about 10%: corporate and captive.** Google (Oct 2024), Asahi India Glass (20.8 MW hybrid, Jan 2026), Ambuja Cements captive blocks (356 MW).
- **Tier 3, 10 to 15%: battery arbitrage.** 3.55 GWh in June 2026 to 6.63 GWh by 1 October, over half of India's 12.6 GWh fleet; Rs 15,000 crore for 10 GWh more in FY27. Charge on curtailed noon solar behind the 9,000 MW cap, discharge 19:00 to 22:00 at the Rs 10/kWh exchange ceiling.

Charge at noon when the line is full, sell at 8 pm when the price hits the ceiling

Stylised day-ahead exchange price and battery dispatch; 6.63 GWh operating at Khavda in October 2026.



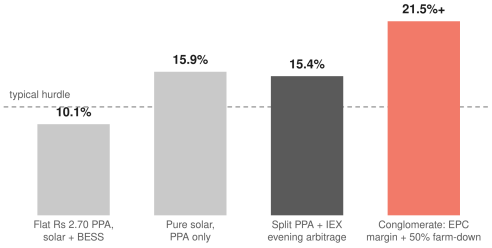
Source: IEX ceiling and AGEL BESS statements; curve shape is illustrative.

Project finance: a 1 GW_{AC} solar + 500 MWh battery SPV

System	1,000 MW_{AC}; 1,350 MWp_{DC} TOPCon on trackers
Storage	250 MW / 500 MWh LFP, 88% round trip
Energy at 33% CUF	2,890.8 MU a year
Capex	**Rs 5,350 crore**
solar, piling, pooling bay	Rs 4,575 crore
battery at Rs 1.55 crore/MWh	Rs 775 crore
Financing, 75:25	Rs 4,012.5 cr debt at 9.0%; Rs 1,337.5 cr equity
Year 1	
base PPA, 2,715.8 MU at Rs 2.70	Rs 733.27 crore
evening arbitrage, 154 MU at Rs 7.80	Rs 120.12 crore
gross revenue; opex	Rs 853.39 crore; Rs 62.00 crore
EBITDA (92.7%)	**Rs 791.39 crore**
debt service; DSCR	Rs 584.05 crore; **1.36**× (1.58× avg)

The SPV alone earns 15%; the group earns 21% or more

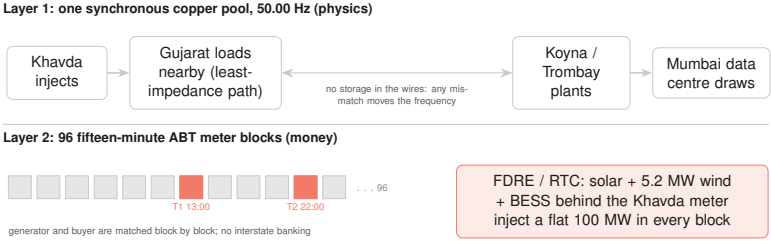
Modelled equity IRR for a 1 GW AC solar + 500 MWh battery SPV, Rs 5,350 crore capex, 75:25 debt.



Source: Analysis note project-finance model; illustrative.

Why the grid is not a free battery: the 15-minute clock

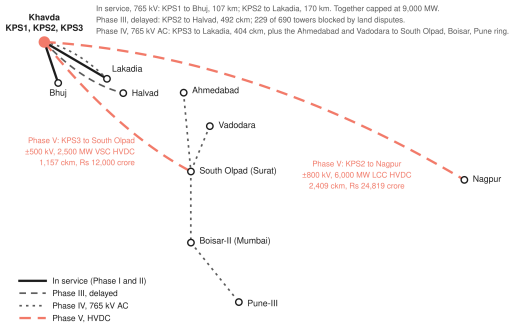
- **Physics.** Electrons from Khavda flow down the least-impedance path to nearby Gujarat loads; a Mumbai data centre draws from nearby Maharashtra plants. Lines store nothing: any mismatch between injection and drawal moves the frequency off 50.00 Hz.
- **Money.** Generator and buyer are metered by 0.2S-class ABT special energy meters in 15-minute blocks, 96 a day. There is no interstate banking, and state rules bar drawing banked solar in the evening peak.
- **So** a flat 100 MW round-the-clock contract needs oversized solar, 5.2 MW wind and batteries *behind* the Khavda meter, injecting 100 MW in every block.



Grid evacuation: pooling stations and the Rs 55,552 crore spine

Two lines out today; a Rs 55,552 crore spine still to come

Evacuation corridors from Khavda. Solid: in service (9,000 MW ceiling). Dashed: under construction.



Source: CTUIL minutes; CMIE CapEx transmission projects; schematic, positions approximate.

Inside the park. 800 V inverters → 33 kV underground collectors → 18 developer pooling substations (33/400 kV) → three 765/400 kV GIS hubs, KPS1 to KPS3 (KPS4 planned).

Phases I and II, in service: the 9,000 MW ceiling. Two 765 kV double circuits: KPS1 to Bhuj (107 km) and KPS2 to Lakadia (170 km).

Phase III, late. KPS2 to Halvad, 492 ckm: 229 of 690 tower sites resisted, Telegraph Act orders, 266.7 ha of forest clearance; December 2025 slipped into 2026/27.

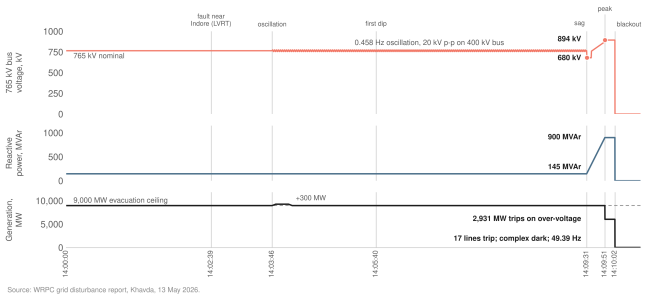
Phase IV, 7 GW AC spine. KPS3 to Lakadia (404 ckm); Vadodara, South Olpad, Boisar-II, Pune-III; 20,000+ MVA of GIS and ±300 to 400 MVAr STATCOMs.

Phase V, 8.5 GW HVDC. ±800 kV, 6,000 MW LCC from KPS2 to Nagpur (2,409 ckm, Rs 24,819 crore); ±500 kV, 2,500 MW VSC from KPS3 to South Olpad (1,157 ckm, Rs 12,000 crore).

Anatomy of the 13 May 2026 blackout (WRPC telemetry)

Dark in 17 seconds: reactive power overshoot in a weak grid

Reconstructed from the WRPC event sequence of 13 May 2026; shapes between events are schematic.



Source: WRPC grid disturbance report, Khavda, 13 May 2026.

Baseline, 14:00. 14,733 MW RE and 1,174 MW BESS connected; 8,963 MW generating, at the 9,000 MW ceiling; 50.00 to 50.06 Hz.

	RE MW	BESS MW	SVG MVAr
KPS1, Adani	7,894	475	4,150
KPS2, NTPC and state	3,155	–	0
KPS3, Adani, NTPC, SRPL	3,684	699	1,900

The cascade. A distant fault near Indore (14:02:39) triggered ride-through; a 0.458 Hz oscillation grew; at 14:09:31 voltage sagged to 680 kV and reactive injection ramped from 145 to 900 MVAr. In a weak grid $\Delta V \approx \Delta Q / S_{sc}$: voltage hit 894 kV, 2,931 MW tripped on over-voltage, lines unloaded and the Ferranti rise worsened. Seventeen lines tripped in 17 seconds; 8,963 MW gone; 49.39 Hz.

Mega-park benchmarking

The largest single-site hybrid complex in the world, and the only one without hydro or coal inertia next door.

	Khavda, Gujarat	Ladakh RE	Talatan, Qinghai	Kubuqi, Inner Mongolia	Bhadla, Rajasthan
Target capacity	30 GW hybrid; 726 km^2	13 GW RE + 12 GWh BESS	19 to 25 GW	16 GW	2.25 GW
Operational, 2026	about 14.8 GW connected; 5.1 GW built in FY26	early works	majority built	phased	complete
Terrain, elevation	sea-level sabkha, 3 to 4 m; brine; seismic zone V	4,500 m passes	dry plateau, 2,900 m, non-corrosive	desert, flexible coal nearby	arid, inland
Grid balancing	LFP batteries, STATCOMs, VSC-HVDC; no synchronous plant nearby	1,268 ckm HVDC, Rs 20,774 crore	1.28 GW Longyangxia hydro, 30 km away	4 GW co-located coal	state grid

Analysis note; public project disclosures. Figures for the comparators are approximate.

Field visit protocol: questions for domain experts

Chief geotechnical and civil engineer

Lateral pile drift and ZAM coating corrosion on the 2024 micro-piles after three monsoon brine cycles?

After 500+ dry-brush passes, how much anti-reflective transmittance is lost to micro-scratching?

How much does construction silt cut rear-side albedo gain against the P90 model?

Industrial strategy lead

What does the filed Rs 6.5 crore per MW include: land, pooling bay, grid connection, storage? Why do sub-park tenants file Rs 3.25 to 5.5?

When Halvad and the HVDC links open, what happens to the evening arbitrage case for the batteries?

Pooling station (KPS1 to KPS3) and SCADA chief

On the live SCADA screen: theoretical generation from pyranometers and anemometers against actual injection and the 9,000 MW ceiling?

Since the 894 kV event of 13 May, how were SVG droop curves retuned, and what share of BESS inverters run grid-forming?

BESS, trading and strategy leads

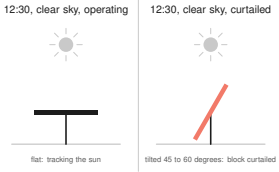
Of yesterday's charge into the 6.63 GWh fleet, what share was otherwise-curtailed power and what share diverted PPA power? Chiller load at 48 °C?

At what GWh scale do evening discharges cannibalise the Rs 10/kWh exchange ceiling?

On-site visual audit and pocket crib sheet

Three things to look at from the car

- **Tracker angles at 12:30.** Rows should lie flat. A block feathered to 45 to 60° under a clear sky is being curtailed.
- **Pile collars.** Where the 2024 piles meet the crust: rust weeping, monsoon scour around the concrete collars.
- **Battery containers at PSS-5, 8, 10.** How many have cooling fans running and HV cables terminated into the PCS skids.



A feathered block at solar noon means the operator is turning it down.

Field equations

Daily energy check. 1,000 MW_{AC} at 33% CUF must export 7.92 MU a day (2,890.8 MU a year). An SPV exporting 2,200 MU a year is running at 25.1%, not 33%: the gap is curtailment.

Cubic wind check. $P \propto v^3$: 8 to 6 m/s is -25% in speed and -57.8% in power.

Weak-grid voltage. $\Delta V \approx \Delta Q / S_{sc}$: with a low short-circuit ratio, excess reactive injection becomes destructive over-voltage. The 13 May peak was 894 kV on a 765 kV system.

Sources